Persistent Scene-Graph Memory on a Legged Robot: Separating Memory Gains from **Execution Failure**

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RoboMem Workshop #53

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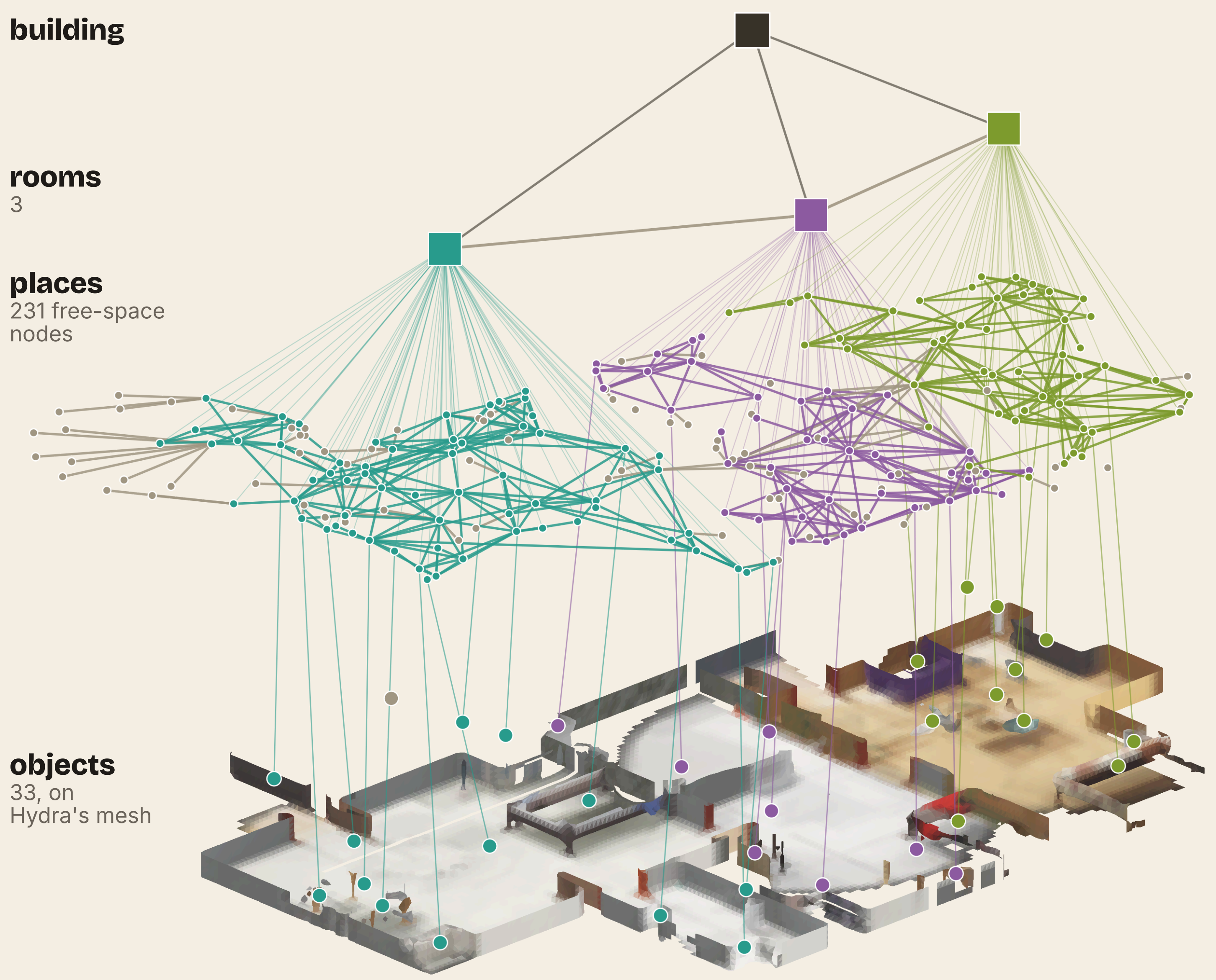
01 THE QUESTION

Does memory still help when the robot can get stuck?

Lifelong navigation benchmarks run in Habitat, where the robot is a cylinder that slides along walls. A Go2 catches on door frames, wedges between furniture and falls. A fall ends every goal left in the house.

02 WHAT THE DOG REMEMBERS



Hydra's 3D scene graph of this house after five goals, carried from goal to goal. Colour marks the room a node belongs to; grey nodes are in no room. SSMG-Nav plans on it.

RGB-D → **Hydra graph** → Qwen3.5-9B scores objects, VLFM frontiers → A* + learned gait

03 THE TEST

A B C A B

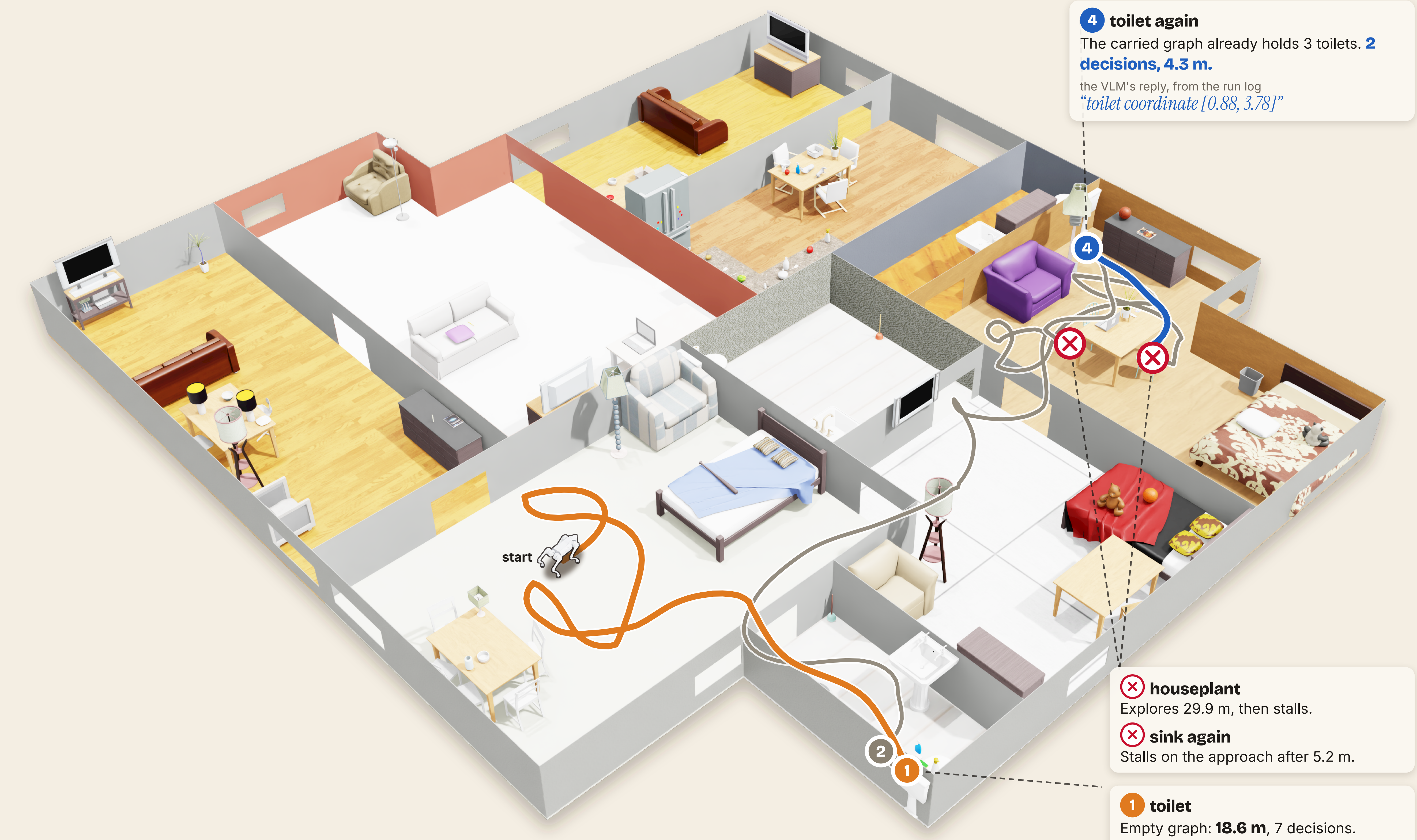
Five object goals per house. Goals 4 and 5 repeat 1 and 2.

	record store off	record store on
graph carried	ssmg-full	ssmg-ukb-full
graph rebuilt	ssmg-blind	ssmg-ukb-blind

25 ProcTHOR houses · 5 repeats · **2,500 subtasks** in Isaac Sim · object labels from the simulator, so perception errors stay out.

04 ONE HOUSE, FIVE GOALS

The second time, it already knows where a toilet is.



4 toilet again
The carried graph already holds 3 toilets. **2 decisions, 4.3 m.**
the VLM's reply, from the run log
"toilet coordinate [0.88, 3.78]"

⊗ houseplant
Explores 29.9 m, then stalls.

⊗ sink again
Stalls on the approach after 5.2 m.

1 toilet
Empty graph: **18.6 m**, 7 decisions.
Goal 2's sink: 1.3 m further.

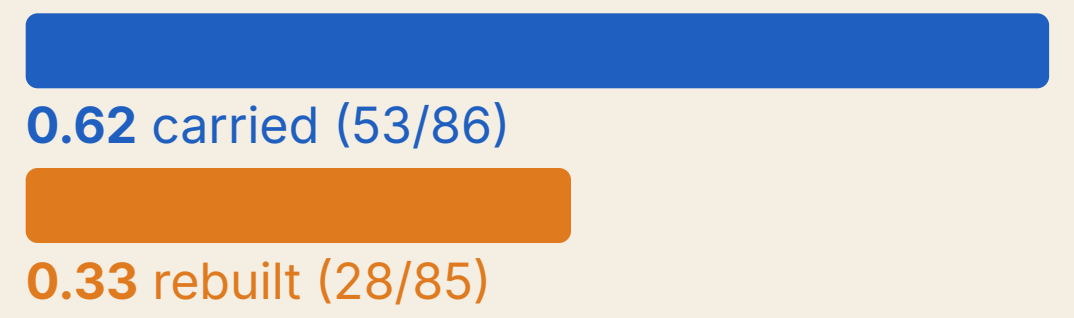
One run from the sweep: house val_15, repeat 5, graph carried. Paths are the logged trajectories, drawn on an Isaac Sim render with the walls cut down.

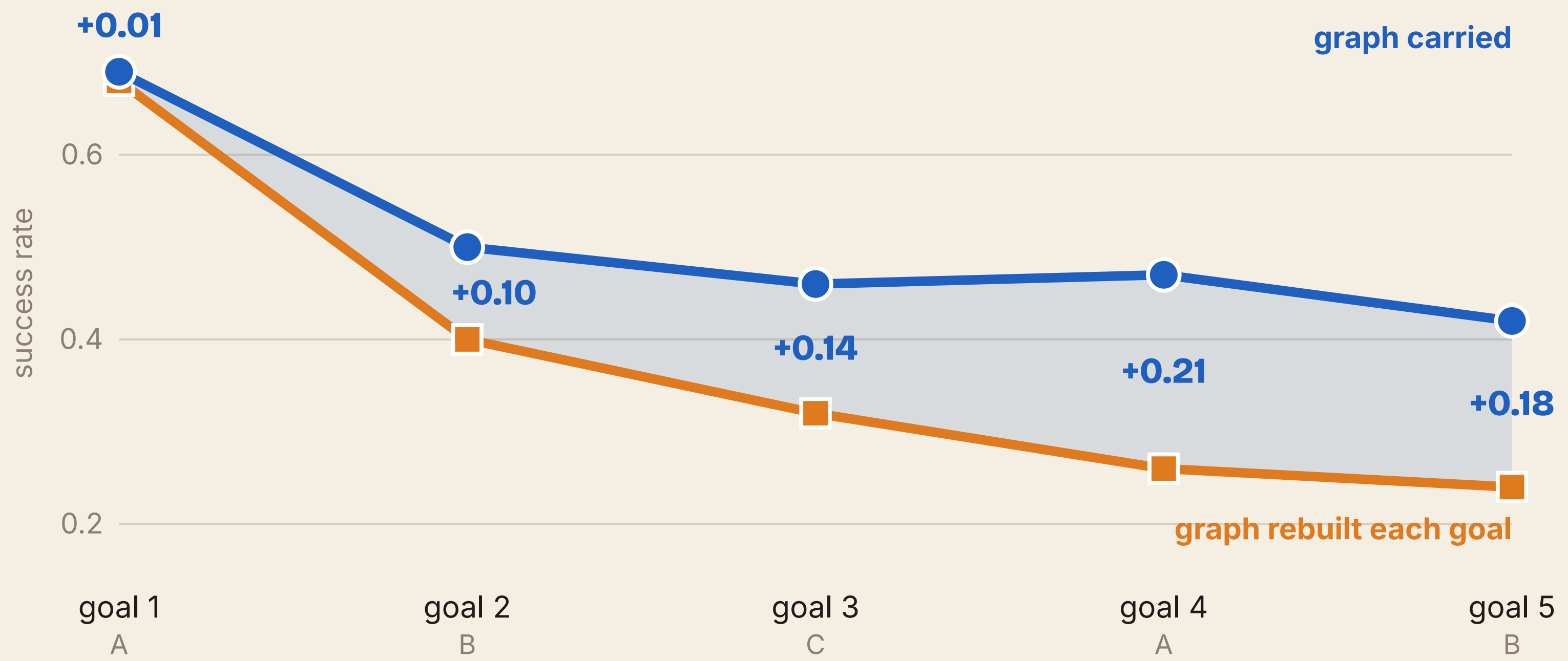
05 ACROSS 2,500 SUBTASKS

0.38→0.51

success rate, graph rebuilt → carried
+0.125 [+0.06, +0.19], $p < 0.001$, 625 pairs

Goal 4 succeeds, if goal 1 did:

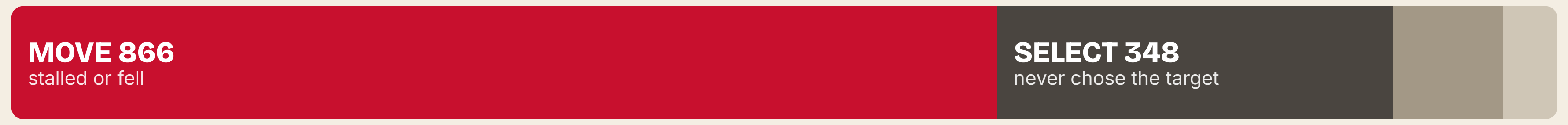




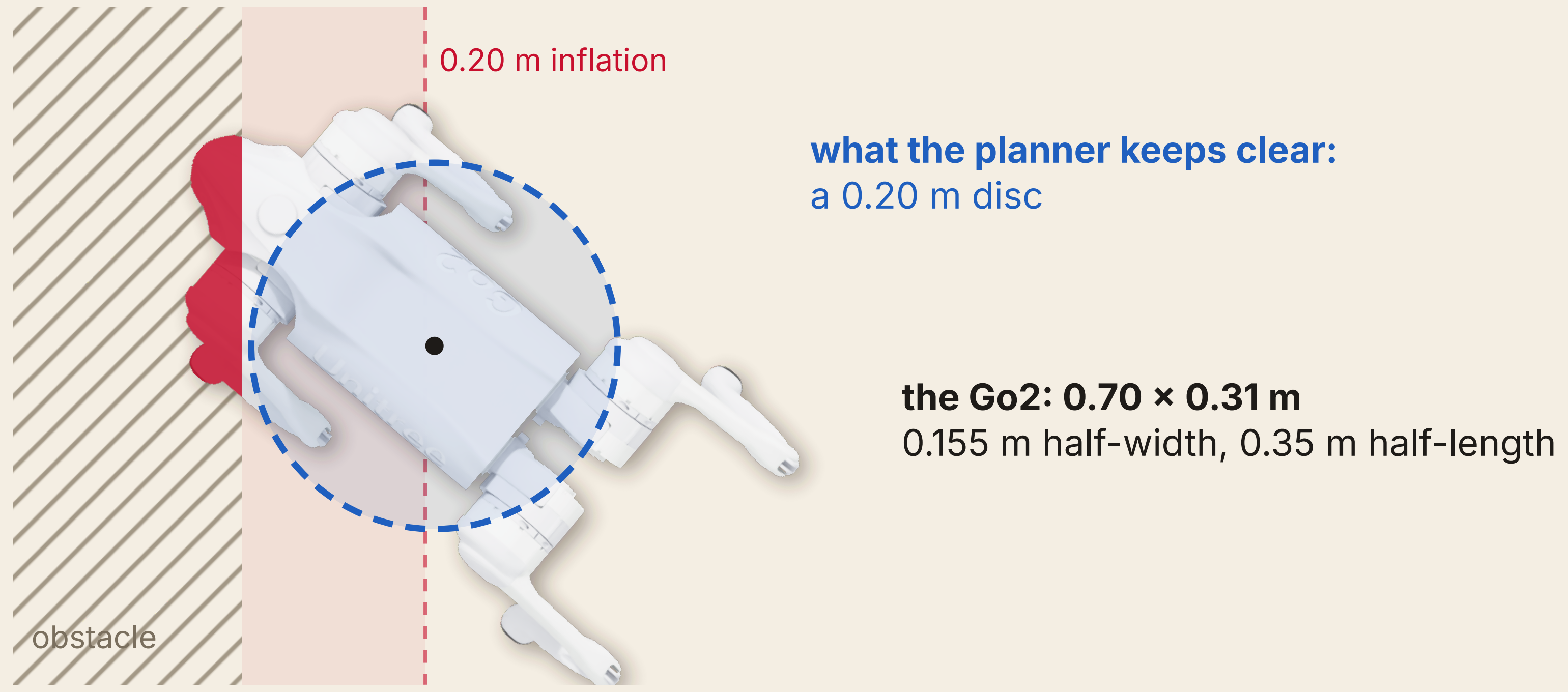
The gain grows as the graph accumulates. Both start goal 1 empty.

06 WHY IT FAILS

64% of 1,357 failures are the robot **stalling or falling**.



also GROUND 97: declared the object absent · no graph 46

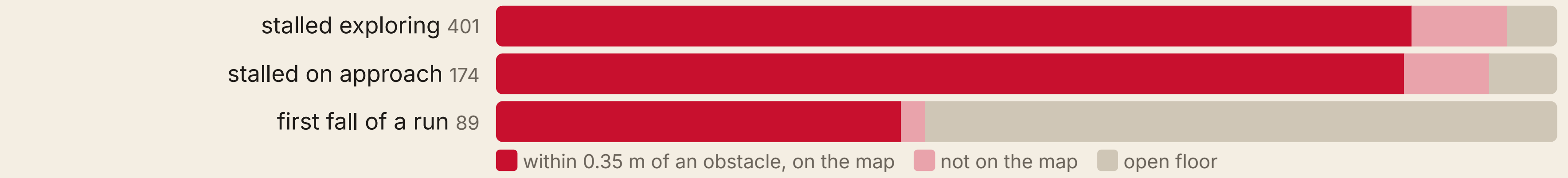


what the planner keeps clear:
a 0.20 m disc

the Go2: 0.70 × 0.31 m
0.155 m half-width, 0.35 m half-length

95% of 575 stalls end within half a body length (0.35 m) of an obstacle. Median clearance: **0.21 m**, right at the planner's 0.20 m inflation. In 86% of stalls the obstacle was already on the map.

WHERE THE ROBOT STOPPED



Falls are different: **53 of 89** happen in open floor, which points to the gait.

07 THE TRAP

Execution failure can fake a memory gain.

	all pairs	both robots decided
record store, graph rebuilt	**+0.051** $p = 0.023$	**+0.021** $p = 0.47$
scene graph carried	**+0.125** $p < 0.001$	**+0.090** $p < 0.001$

The record store's 32 extra successes come with 33 fewer subtasks lost before the first decision.

Takeaway. Memory evaluations on legged robots should report execution-failure rates for every configuration, and the effect on pairs in which both robots made a decision.

Limitations: one simulator, one gait policy, one house generator; labels from the simulator; open-loop recovery, no contact logging, so obstacle proximity is evidence, not proof; no ideal-body baseline.

Hydra (Hughes et al., RSS 2022) · SSMG-Nav (Niu et al., 2026) · GOAT-Bench (Khanna et al., CVPR 2024) · VLFM (Yokoyama et al., ICRA 2024) · ProcTHOR (Deitke et al., NeurIPS 2022)